Low-Rank Compression of Neural Network Weights by Null-Space Encouragement

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***Abstract*—Deep neural networks are often heavily over-parametrized for their purposes, making them costly to store and run on resource-constrained devices. Therefore, compressing them into fewer parameters with minimal performance loss is essential. One popular way is the low-rank decomposition because high-dimensional data often lie near a lower-dimensional manifold, and the most important features can be captured by the dominant singular values of the weight matrices. In this work, we first compress pretrained models into their low-rank forms (SVD or CP). The compressed low-rank network becomes the student and the original over-parametrized network becomes the teacher. Next, we exploit the low-rank structure of the approximated weight matrices by projecting the mismatch of intermediate activations onto the null space of the low-rank weight matrices, ensuring that the student focuses on the teacher's task-relevant directions. Finally, we use knowledge distillation to align outputs of the student and the teacher. We validate our approach on tasks involving fully connected layers and/or convolutional layers, achieving up to 90% parameter and 80% FLOP reduction, with either $< 1\%$ accuracy drop or up to 1.68% gain on MNIST, CIFAR-10, and CIFAR-100 benchmarks. We also compare our method with previously proposed FitNets [1] and Gramian [2] to show that the proposed technique is better at maintaining higher accuracy. These results indicate that the proposed technique not only reduces model size but also acts as a regularizer, improving generalization.**

I. Introduction

Deep neural networks have shown revolutionary results in various domains from image recognition to natural language processing, emerging from the success of [3] and later of [4].

2) We propose to employ *null-space absorbing loss* in the supervised fine-tuning stage of the compression to encourage the student to conduct parameter updates such that the difference between resulting hidden activations of the student and the original activations of the teacher are kept within the kernels of the teacher weights. This ensures only task-relevant directions are transferred to the student rather than point-to-point matching.
3) We aim for accuracy-focused compression rather than merely parameter reduction and incorporate knowledge distillation to fine-tune the student, enabling about 80% FLOP reduction keeping about the same performance on MNIST, CIFAR-10 and CIFAR-100.

The remainder of the paper is organized as follows. Section II reviews related work on neural network compression. Section III details our proposed framework. Section IV presents our experimental results, and Section V concludes with future directions.

II. Related Work

In [6], *knowledge distillation* is introduced by training a compact student network from a larger teacher network. Specifically, the student minimizes the Kullback-Leibler (KL) divergence between its softened output probability distributions and the teacher's, which is controlled with a temperature hyperparameter. This foundational technique proved that compact models can approximate larger models with minimal perfor-